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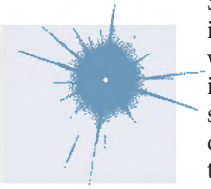
315 Sepsis and Septic Shock

Jeffrey R. Strich, Daniel S. Chertow



Sepsis is an infectious syndrome that results in considerable morbidity, mortality, and long-term sequelae among survivors. Sepsis, derived from the Greek word *sipsi*, "to make rotten," was first described by Hippocrates in the 400s B.C. In the 1800s, Sir William Osler opined on sepsis by saying, "except on few occasions, the patient appears to die from the body's response to infection rather than from it." The first consensus definition of sepsis, published in 1992, recognized sepsis as the body's systemic response to infection. To operationalize this definition, systemic inflammatory response syndrome (SIRS) clinical criteria were established, which included temperature <36°C or >38°C, heart rate >90 beats/min, respiratory rate >20 breaths/min or partial pressure of CO_2 <32 mmHg, and leukocyte count <4000/μL or >12,000/μL or >10% bands. Suspected infection with two or more SIRS criteria was classified as sepsis, while the term *severe sepsis* required evidence of hypoperfusion or end-organ dysfunction including oliguria, acute alteration in mental status, or lactic acidosis. Septic shock was defined as sepsis-induced hypotension, determined by systolic blood pressure <90 mmHg or reduced ≥40 mmHg from baseline, absent other causes, despite adequate volume resuscitation. Sepsis-2, the second consensus definition established in 2003, acknowledged clinical complexity beyond existing SIRS criteria and expanded the list of clinical and laboratory criteria to diagnose sepsis. Sepsis-3, established in 2016 as the current consensus definition, abandoned SIRS criteria, given poor specificity for distinguishing sepsis from other noninfectious inflammatory processes, and eliminated the term *severe sepsis*. Sepsis was redefined as life-threatening organ dysfunction caused by a dysregulated host response to infection, and septic shock was defined as a subset of sepsis in which profound circulatory, cellular, and metabolic abnormalities increase mortality risk beyond sepsis alone.

EPIDEMIOLOGY

Each year, an estimated 48.9 million sepsis cases occur globally, and 1.7 million cases occur in the United States. Data from >400 U.S. academic, community, and federal hospitals indicate that sepsis occurs in ~6% of hospitalized adults, with stable incidence over time **(Fig. 315-1)**. Approximately 11 million sepsis deaths occur globally each year, accounting for one in five total global deaths, with 85% occurring in low- and middle-income countries. An estimated 350,000 sepsis deaths or discharges to hospice occur annually in the United States, with an estimated 15% overall mortality among U.S. hospitalized adults with sepsis (Fig. 315-1) and up to 40% mortality in patients with septic shock. Data from the U.S. Centers for Medicare and Medicaid Services indicate the aggregate cost for inpatient hospital and skilled nursing facility sepsis admissions was an estimated $13.4 billion dollars in 2018. A retrospective evaluation of sepsis costs in U.S. hospitals during 2010–2016 suggests a cost of $16,324 and $38,298 per sepsis and septic shock admission, respectively.

PATHOGENS AND SITES OF INFECTION

Approximately 88% of sepsis cases are community onset, defined as being detected within 48 h of hospitalization, whereas an estimated 12% are hospital onset, detected after 48 h of hospitalization. While viruses, fungi, and other pathogens may induce sepsis, the role of bacterial infection is best described. An estimated 53% of sepsis cases in the United States are bacterial culture positive, with a relatively even split between gram-positive and gram-negative organisms.

A retrospective cohort study of 17,430 adult community-onset culture-positive sepsis cases from 104 U.S. hospitals during 2009 to 2015 identified *Staphylococcus aureus*, *Streptococcus* spp., and *Enterococcus* spp. as the most prevalent gram-positive organisms isolated, of which 13.6% were antibiotic-resistant, including methicillin-resistant *S. aureus* (MRSA) and vancomycin-resistant enterococci (VRE). *Escherichia coli*, *Klebsiella* spp., and *Pseudomonas aeruginosa* were the most frequently isolated gram-negative organisms, of which 13.2% were resistant to ceftriaxone, extended-spectrum β-lactams, or carbapenems. The most frequently reported anatomic site of primary infection was the urinary tract (48.9%), followed by the respiratory tract (32.9%), an intraabdominal site (13.6%), and skin or soft tissue (10.3%). Bacteria were most frequently isolated from urine (52.1%), blood (40.0%), and the respiratory tract (16.7%).

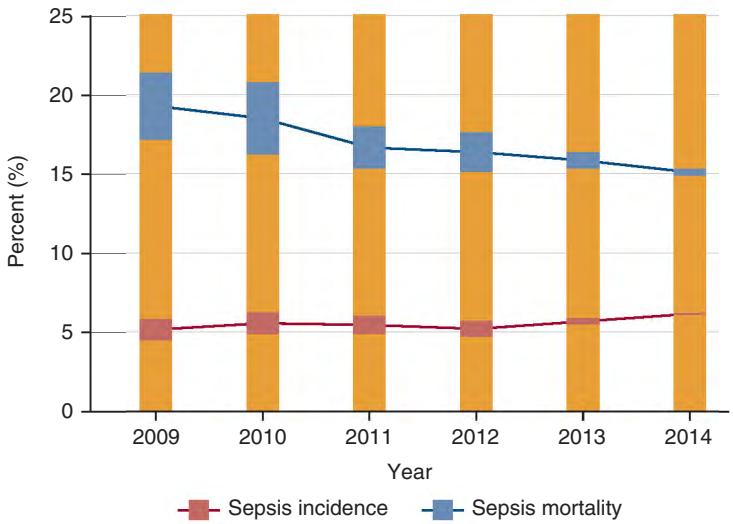


FIGURE 315-1 Sepsis incidence and mortality. Sepsis incidence and mortality. Sepsis incidence is expressed as a proportion of sepsis cases among 7,801,624 adult hospitalizations across 409 U.S. hospitals from 2009–2014. Mortality is the proportion of sepsis deaths among sepsis cases.

Patients with hospital-onset sepsis differ from community-onset sepsis patients in that they more often have comorbidities and experience intraabdominal infections and bacteremia with *Enterococcus*, *Candida*, and *Pseudomonas*. Patients with hospital-onset sepsis also experience higher intensive care unit (ICU) admission rates, longer hospital length of stay, and increased mortality.

RISK AND PROGNOSTIC FACTORS

U.S. Centers for Disease Control and Prevention (CDC) data from 2021 show that sepsis mortality increases with age and is higher among men than women in all age groups: 232.7 versus 173.0 (65–74 years), 477.3 versus 349.8 (75–84 years), and 1037.8 versus 755.5 (≥85 years) deaths per 100,000 population, respectively. Preexisting medical conditions including diabetes and obesity; neurologic, respiratory, or cardiac conditions; renal or hepatic insufficiency; and cancer or other immunosuppressing conditions increase sepsis mortality risk. Recent hospital admission for any reason has also been associated with threefold increased risk of developing sepsis in the following 90 days.

Multiple composite illness severity scoring systems have been applied to hospitalized septic patients to predict outcome or guide stratification or post-hoc analyses in clinical trials or observational studies of sepsis. For example, the Sequential Organ Failure Assessment (SOFA) score quantifies dysfunction in six organ systems including neurologic, using the Glasgow Coma Scale score; cardiovascular, using mean arterial blood pressure or use of vasoactive agents; respiratory, using the ratio of arterial blood partial pressure of oxygen (Pa_{O_2}) to fraction of inspired oxygen (F_{IO_2}) or use of mechanical ventilation; hepatic, using serum bilirubin levels; renal, using serum creatinine levels; and coagulation, using blood platelet levels. Elevated SOFA scores are associated with increased sepsis mortality at the population level but cannot accurately predict outcomes of individual patients.

Improved stratification of sepsis patients based on clinical "phenotype," determined by clustering of clinical variables using machine learning tools to better predict outcome, has been proposed. In a retrospective analysis of >45,000 patients who met Sepsis-3 criteria, statistical, machine learning, and simulation tools were applied to clinical data obtained within 6 h of emergency room presentation to identify α, β, γ, and δ phenotypes, with associated 2%, 5%, 15%, and 32% mortality rates, respectively. Beyond using immediately available clinical data to identify sepsis phenotypes, integration of transcriptional and proteomic data to define sepsis "endotypes" has also been attempted. While these approaches hold promise to guide improved clinical trial design and patient care, additional research is needed to contextualize and operationalize them across patients and pathogens.

PATHOGENESIS

The host-pathogen interaction during sepsis is heterogeneous, based on patient demographic and clinical factors and pathogen type and virulence. Consequently, the pathophysiology and molecular pathogenesis of sepsis vary across hosts and pathogens. However, a working framework for sepsis pathogenesis, with commonalities across hosts and pathogens, provides a basis for understanding which factors contribute to organ injury and dysfunction during severe infections. During local infection, a prototypical physiologic response is characterized by pathogen recognition followed by balanced inflammatory, anti-inflammatory, and repair responses resulting in pathogen clearance with minimal disruption to systemic homeostasis. During sepsis, however, pathogen components and exuberant cellular and soluble immune responses contribute to systemic illness, resulting in end-organ injury and dysfunction **(Fig. 315-2)**. Pathogen components and host responses to infection may also impair or delay adaptive immunity and tissue repair.

Pathologic responses in sepsis are mediated by myeloid lineage cells (i.e., neutrophils, monocytes, macrophages, and dendritic cells), lymphoid lineage cells (i.e., natural killer [NK] cells and lymphocytes), parenchymal cells (e.g., endothelial and epithelial cells), and soluble mediators (e.g., cytokines, chemokines, nitric oxide, histamine, prostaglandin, and bradykinin). Activated leukocytes adhere to endothelium, migrate into tissue, and perform end-effector functions, including reactive oxygen species (ROS) generation, that contribute to tissue injury. Activation of endothelial cells, platelets, the clotting cascade, and the complement system contribute to a prothrombotic state. When platelets and clotting factors are consumed, hemorrhage risk increases. Pathophysiologic manifestations of sepsis include systemic vasodilation, capillary leakage with interstitial fluid accumulation, and microvascular thrombosis resulting in impaired oxygen delivery, uptake, and utilization. Taken together, these factors drive cellular injury and end-organ dysfunction.

TRIGGERS OF HOST IMMUNE RESPONSES

Host immune responses in sepsis are initiated by pathogen components termed pathogen-associated molecular patterns (PAMPs) and propagated by host factors termed damage-associated molecular patterns (DAMPs). PAMPs and DAMPs interact with innate and adaptive immune cells and parenchymal cells early during sepsis, amplifying immune responses. PAMPs include bacterial lipopolysaccharide (LPS) or other bacterial cell wall or membrane components, pathogen nucleic acids, including single- or double-stranded RNA, and other pathogen-related molecules. DAMPs include proteins, lipids, nucleic acids, and other components released from injured cells. Examples include histones, high mobility group box 1 proteins, S100 proteins, oxidized phospholipids, double-stranded DNA, and adenosine triphosphate. PAMPs and DAMPs are recognized by extra- or intracellular receptors, termed pattern recognition receptors (PRRs). Multiple PRRs, including toll-like receptors (TLR), of which 10 have been described in humans, C-type lectin receptors (CLR), receptors for advanced glycation end products (RAGE), retinoic acid–inducible gene-I-like receptors (RIG-I), and nucleotide-binding oligomerization domain-like (NOD) receptors, have been implicated in sepsis signaling. Recognition of PAMPs and DAMPs by PRRs on innate immune cells such as neutrophils, monocytes, and macrophages activates inflammatory pathways, triggers release of proinflammatory cytokines and chemokines, and increases surface expression of vascular adhesion molecules.

MYELOID RESPONSES

Neutrophils, monocytes, and macrophages provide early innate immune defenses against invading pathogens but also contribute to sepsis pathogenesis by promoting inflammation and cellular injury and, in some cases, limiting adaptive immunity. Differentiated mature neutrophils are the most abundant circulating leukocytes in healthy individuals. Upon activation during sepsis, neutrophils upregulate surface adhesion molecule expression (e.g., CD11b) to bind endothelium. Within the microvasculature, neutrophils release neutrophil extracellular traps (NETs), web-like structures of DNA decorated with antimicrobial proteins such as cathepsin-G, myeloperoxidase, and neutrophil elastase to limit dissemination of invading pathogens. NETs interact with activated platelets, endothelial cells, and fibrin to form microvascular thromboses. Activated neutrophils also migrate into tissues to combat microbes through phagocytosis, degranulation, ROS generation, NET formation, and proinflammatory cytokine and chemokine release. Neutrophil granule components degrade extracellular matrix, while ROS oxidize proteins and lipids contributing to cellular injury and dysfunction. Soluble inflammatory mediators further recruit immune cells to sites of infection, exacerbating inflammation and cellular injury. Mediators such as interleukin (IL) 6, granulocyte-macrophage colony-stimulating factor, and granulocyte colony-stimulating factor trigger release of immature neutrophils from the bone marrow, a process termed emergency granulopoiesis that is associated with poor outcomes in septic patients.

Monocytes are innate myeloid cells that circulate in healthy individuals for up to 7 days following release from the bone marrow. Three main monocyte populations are defined by CD14 and CD16 surface expression, including classical (CD14+, CD16–), intermediate (CD14+, CD16+), and nonclassical ($CD14^{low}CD16+$) monocytes. Classical monocytes account for 85% of circulating monocytes in healthy individuals but rapidly migrate to infected tissues during sepsis, resulting in transient monocytopenia and an increased proportion of circulating nonclassical and intermediate monocytes. Nonclassical monocytes are more terminally differentiated blood-resident monocytes thought to patrol the endothelium during sepsis. Monocytes

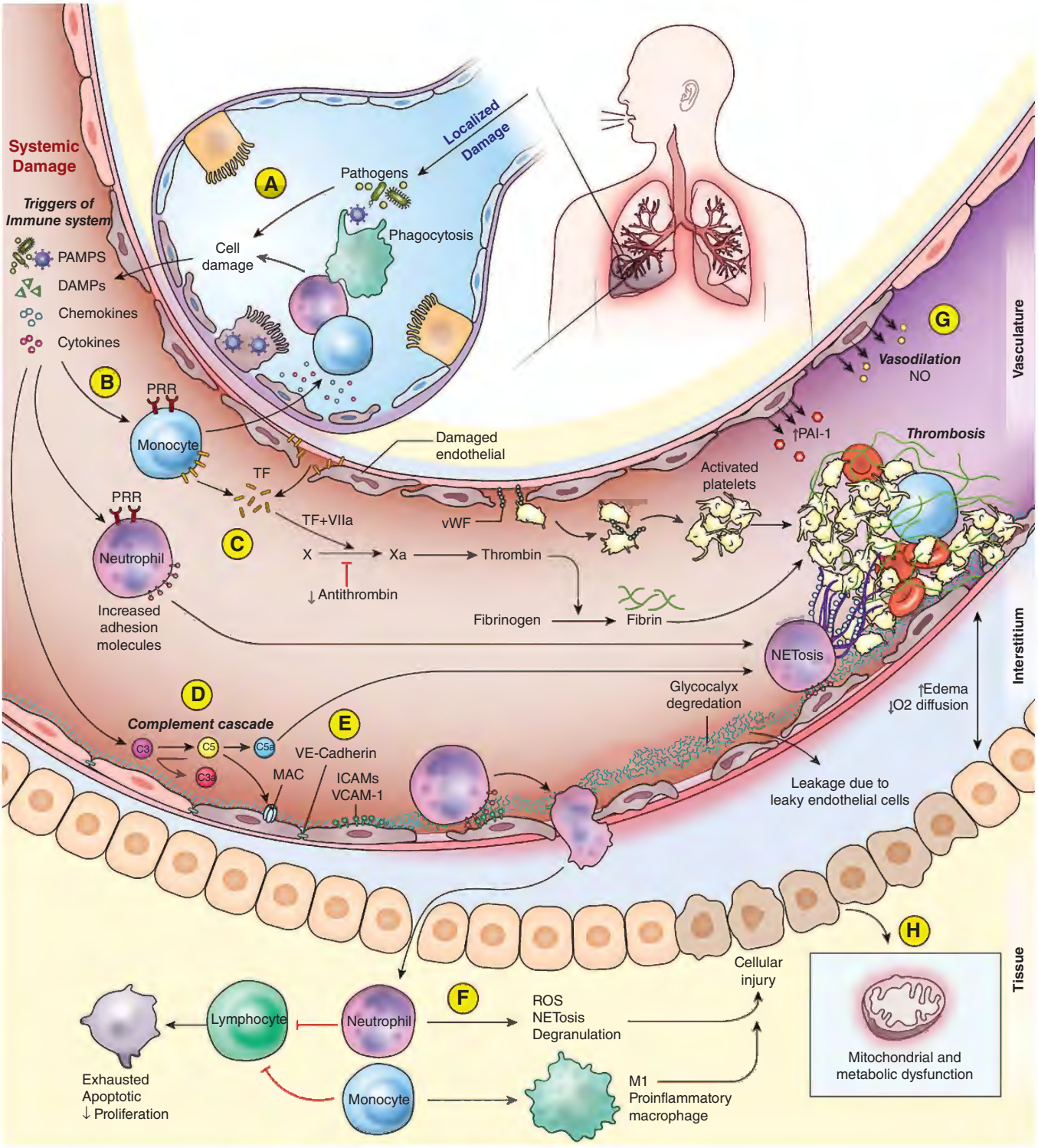


FIGURE 315-2 Sepsis pathogenesis. (A) Invading pathogens and tissue-resident and recruited leukocyte responses induce localized cell damage. **(B)** Systemic damage is mediated by pathogen-associated molecular patterns (PAMPs), damage-associated molecular patterns (DAMPs), cytokines, and chemokines that activate pattern recognition receptors (PRRs) on circulating monocytes and neutrophils. **(C)** Increasing tissue factor (TF) on endothelium and leukocytes triggers the clotting cascade and fibrin generation. Adherent neutrophils release extracellular traps (NETs) and microthrombi form, composed of leukocytes, fibrin, and platelet aggregates, bound together by von Willebrand factor (vWF). Antithrombin levels are decreased, promoting clot formation, and plasminogen activator inhibitor-1 (PAI-1) levels are increased, impairing clot breakdown. **(D)** Complement activation generates C3a and C5a, which activate platelets and myeloid cells. The C5b-9 membrane attack complex (MAC) promotes endothelial injury. **(E)** Activated and injured endothelium increase intercellular adhesion molecules (ICAMs) and vascular cell adhesion molecule (VCAM)-1 expression, glycocalyx breakdown, apoptosis, and loss of intercellular vascular endothelial (VE)-cadherin and tight junctions contributing to interstitial edema, decreased oxygen diffusion, and leukocyte migration. **(F)** Activated neutrophils degranulate and release reactive oxygen species (ROS) and NETs, and monocytes differentiate into proinflammatory (M1) macrophages, contributing to cellular injury and dysfunction. Activated myeloid cells also suppress lymphocyte function, through exhaustion, apoptosis, and decreased proliferation. **(G)** Arterial vasodilation, partially mediated by excess nitric oxide, contributes to tissue hypoperfusion and ischemia. **(H)** With or without adequate oxygen delivery, cellular mitochondrial and metabolic functions are disrupted during sepsis.

that migrate into tissues are exposed to PAMPs, DAMPS, cytokines, and other mediators, stimulating them to differentiate into dendritic cells or macrophages. Dendritic cells are phenotypically and transcriptionally diverse professional antigen-presenting cells with a broad PRR and cytokine secretion repertoire. These cells contribute to pathogen clearance and inflammation and bridge innate and adaptive immune responses through antigen presentation and lymphocyte activation. Macrophages that originate from bone marrow–derived monocytes are distinct from tissue-resident macrophages, which derive from embryonic progenitors. Depending on environmental conditions within tissues, bone marrow–derived monocytes polarize to become proinflammatory (M1) or anti-inflammatory, reparative (M2) macrophages. M1 macrophages contribute to pathogen clearance through phagocytosis, extracellular trap formation, ROS generation, and inflammatory cytokine release (e.g., IL-1β, IL-6, and tumor necrosis factor [TNF]-α). However, these same effector functions contribute to tissue injury. Macrophages also process and present antigens on surface major histocompatibility complex (MHC) II molecules, which are required for development of antigen-specific effector and memory B- and T-cell responses. This essential link between myeloid and lymphoid cells can be disrupted by diverse pathogen-induced mechanisms including downregulation of MHC surface expression or interruption of antigen processing or presentation by pathogen proteins and other components.

While myeloid cells typically contribute to inflammation in sepsis, a subset of pathologically activated myeloid cells in tissues, termed myeloid-derived suppressor cells, functionally suppress NK-cell, T-cell, and B-cell function, contributing to immunosuppression in sepsis.

LYMPHOID RESPONSES

NK cells are a heterogenous group of innate lymphocytes that migrate to sites of infection, contribute to pathogen clearance, and mediate inflammatory responses during sepsis. NK cells have activating and inhibitory cell surface receptors that modulate their activity. NK-cell PRRs recognize PAMPs including LPS, peptidoglycan, and double-stranded RNA. NK-cell cytokine receptors recognize inflammatory cytokines including type 1 interferons (IFN) and IL-12. Activated NK cells release proinflammatory cytokines including IFN-γ and IL-32, which activate myeloid cells in a positive feedback loop. Activated NK cells also directly kill infected cells that have downregulated MHC class I surface molecule expression, through release of cytotoxic granules including perforin and granzyme. In studies of human sepsis, circulating NK cells with increased CD69 expression, indicative of activation, and increased plasma concentrations of IFN-γ and granzyme A and B support a proinflammatory role of NK cells. However, NK cells also release IL-10, known to suppress myeloid cell–mediated inflammatory responses, highlighting an anti-inflammatory role of NK cells in sepsis as well.

B- and T-cell responses in sepsis are essential for recognizing and clearing pathogens. B cells produce antibodies against foreign antigens and form antigen-specific memory cells. CD8+ T cells lyse infected cells that present foreign antigens in association with MHC surface molecules and form memory cells. CD4+ (T_H1) cells activate CD8+ T cells and support memory T-cell formation, CD4+ (T_H2) cells contribute to B-cell class switching, and CD4+ (T_H17) cells protect against extracellular fungal and bacterial infections. B- and T-cell dysfunction in sepsis impairs adaptive immunity and predisposes to subsequent infections among survivors. Contributing factors to adaptive immune dysfunction in sepsis include death of lymphocytes in the circulation and tissue, cellular exhaustion, decreased proliferation, and apoptosis-refractory regulatory T cells, all of which can inhibit inflammatory responses. Functional manifestations of T-cell exhaustion include impaired cytokine production and diminished cytotoxic activity, while diminished antibody production is observed in exhausted B cells.

ENDOTHELIAL ACTIVATION

During homeostasis, endothelial cells line blood vessels and regulate vascular tone and the exchange of cells, fluid, and molecules between the bloodstream and surrounding tissues. The endothelial glycocalyx, a network of proteoglycans, glycoproteins, glycolipids, glycosaminoglycans, and bound plasma proteins, lines luminal endothelium and maintains homeostatic functions. During sepsis, PAMPs and DAMPs activate endothelial cells, altering their structure and function. Glycocalyx breakdown disrupts blood flow and mediates proadhesive, procoagulant, and proinflammatory endothelial properties. Activated endothelial cells increase surface adhesion molecule expression, including intercellular adhesion molecules (ICAMs) and vascular cell adhesion molecule (VCAM)-1, that coordinates leukocyte adhesion, rolling, and diapedesis. Activated endothelium promotes clotting through increased tissue factor (TF) expression and platelet adhesion and impairs fibrinolysis through increased plasminogen activator inhibitor (PAI)-1 release, among other factors. Activated endothelial cells also produce proinflammatory cytokines, including IL-1β, IL-6, and IFN-α, contributing to local and systemic inflammation. Beyond glycocalyx breakdown, barrier function is impaired through endothelial cell apoptosis and loss of intercellular adherence (e.g., vascular endothelial [VE]-cadherin) and tight junctions. Loss of endothelial barrier function contributes to interstitial fluid accumulation, increased interstitial pressure, and tissue hypoperfusion. Alterations in endothelial cell nitric oxide (NO) metabolism in sepsis contribute to systemic vasodilation and cellular injury.

COAGULOPATHY

The coagulation system, composed of plasma proteins, platelets, and the endothelium, maintains vascular integrity. During sepsis, PAMPs and DAMPs interact with these components, innate immune cells, and the complement system to promote thrombus formation and impair clot breakdown. Increased TF expression on activated leukocytes and endothelium triggers the extrinsic clotting cascade, thereby driving thrombin-mediated conversion of fibrinogen to fibrin. Endothelial injury exposes von Willebrand factor (vWF) on subendothelial surfaces, which binds platelets via glycoprotein Ib receptors, and platelets bind fibrinogen via glycoprotein IIb/IIIa receptors. Activated platelets and endothelium release additional vWF from intracellular stores (i.e., Weibel-Palade bodies in endothelial cells), forming multimers that aggregate platelets at the endothelial surface. Activated monocytes and neutrophils bind to these platelet aggregates, and neutrophils release NETs, contributing to a meshwork of fibrin-rich microthrombi. The complement system, which is composed of >50 soluble or membrane bound proteins, also contributes to microthrombi formation. During sepsis all three complement pathways (classical, alternative, and lectin) become activated. Release of C3a and C5a anaphylatoxins further activates platelets and myeloid cells, and assembly of the membrane attack complex (MAC), composed of C5b-9, promotes endothelial injury. Counterregulatory anticoagulant molecules, including antithrombin, tissue factor pathway inhibitor, and activated protein C, are impaired in sepsis, and increased activity of antifibrinolytic molecules, including PAI and thrombin-activatable fibrinolysis inhibitor, limit clot breakdown. When clotting factors and platelets are consumed, spontaneous and provoked hemorrhage can occur.

CELLULAR INJURY AND DYSFUNCTION

A hallmark of sepsis is multiorgan cellular injury and dysfunction. Arterial hypotension due to systemic vasodilation, myocardial depression, and hypovolemia; increased interstitial edema due to vascular leakage resulting in increased oxygen diffusion distance; and microcirculatory dysfunction in part due to endothelial activation, injury, and microvascular thrombosis all contribute to inadequate oxygen delivery and cellular injury and dysfunction during sepsis. Cell death or dysfunction can also result from direct interactions with PAMPs, DAMPs, activated leukocytes, and their associated mediators including cytokines, chemokines, toxic granules, ROS, and reactive nitrogen species (RNS), including peroxynitrite, nitrogen dioxide, and dinitrogen trioxide. ROS and RNS contribute to protein and lipid oxidation and DNA damage of proximate cells. Even in the setting of adequate oxygen delivery, cellular mitochondrial and metabolic functions are disrupted during sepsis, including ATP generation through glycolysis and oxidative phosphorylation, ion (i.e., Na^+, K^+, and Ca^{2+}) homeostasis, and regulation of cell death pathways.

CLINICAL MANIFESTATIONS AND MANAGEMENT

Sepsis may present with nonspecific signs and symptoms including fever, tachycardia, lethargy, and myalgias with or without localizing end-organ findings such as cough, pyuria, or abdominal pain or evidence of end-organ dysfunction such as oliguria or altered mental status. There is no gold standard diagnostic test for sepsis, so epidemiologic, demographic, clinical, laboratory, radiologic, and microbiologic parameters must be considered in diagnosing sepsis. In addition to a focused history and physical examination to elicit signs, symptoms, and physical manifestations of sepsis, an initial laboratory evaluation should include complete blood count with differential, basic metabolic panel, liver function test, serum lactate, coagulation panel, urinalysis, and point-of-care pathogen testing when available. Microbiologic testing with culture of blood and other potentially infected sites (e.g., urine, sputum, wound), ideally prior to antimicrobial administration, should be performed to identify a specific infection and guide antimicrobial therapy. Focused imaging examination by x-ray, computed tomography, or ultrasound of potentially infected sites (e.g., lung, abdomen) to support a diagnosis of sepsis and guide efforts to control the source of infection should also be pursued. Sepsis mimics include but are not limited to noninfectious inflammatory febrile syndromes such as connective tissue diseases and vasculitides; heart failure and other noninfectious causes of pneumonitis or lung injury; noninfectious abdominal syndromes including mesenteric ischemia and inflammatory bowel disease; and hypotension due to noninfectious causes including hypovolemia, autonomic dysfunction, and adrenal insufficiency. In a recent study of septic patients admitted to the ICU, 25% were retrospectively deemed to have sepsis mimics. The most common mimics were cardiovascular (e.g., heart failure, cardiac arrest) and respiratory (e.g., noninfectious chronic obstructive pulmonary disease). In patients with suspected sepsis or septic shock in whom infection is not confirmed, continuous reevaluation for alternative diagnoses is imperative.

RECOGNIZING SEPSIS

Early recognition and treatment of bacterial septic shock with appropriate antibiotics has been associated with reduced mortality. Therefore, the 2023 CDC Hospital Sepsis Program Core Elements guidance document and the 2021 Surviving Sepsis Campaign guidelines recommend hospitals have dedicated sepsis improvement programs that include standard operating procedures for sepsis screening and early treatment. To operationalize updated Sepsis-3 definitions, consensus criteria for sepsis include an increase of ≥2 in SOFA score from baseline in patients with suspected or confirmed infection, and criteria for septic shock include septic patients requiring vasopressor therapy to maintain a mean arterial pressure >65 mmHg and lactate >2 mmol/L despite fluid resuscitation. To more rapidly screen patients for sepsis, the quick (q)SOFA score has been proposed, which includes respiratory rate ≥22 breaths/min, Glasgow Come Scale score <15, and systolic blood pressure ≤100 mmHg. A qSOFA score ≥2 is associated with poor outcome and is more specific, although less sensitive, than SIRS criteria for identifying patients with end-organ dysfunction due to infection. Multiple other sepsis screening tools have been proposed including the National Early Warning Score, the Modified Early Warning Score, and the artificial intelligence–based Targeted Real-Time Early Warning System. Given that each screening tool has advantages and limitations, none is preferentially endorsed for sepsis screening by the 2021 Surviving Sepsis Campaign guidelines.

INITIAL SEPSIS MANAGEMENT

Initial treatment of patients with sepsis or septic shock includes timely blood pressure and end-organ support, antimicrobial therapy, and identification and elimination of the source of infection. The 2021 Surviving Sepsis Campaign treatment guidelines provide the most up-to-date and evidenced-based approach for initial and subsequent treatment of patients with sepsis and septic shock. Initial management recommendations include obtaining intravenous access with peripheral or central venous catheters, administering appropriate and timely antibiotics, treating life-threating hypotension with intravenous crystalloid and vasopressor therapy, and managing respiratory insufficiency or failure with supplemental oxygen, airway support, and mechanical ventilation, when indicated. Following initial stabilization in patients who are critically ill or in shock, admission to the ICU within 6 h should be targeted.

Of these interventions, early appropriate antibiotic administration in patients with bacterial septic shock has been most clearly associated with improved survival. In patients with bacterial septic shock, there is an estimated 7–8% increase in mortality for every 1-h delay in appropriate antibiotic administration following shock recognition. Consequently, in patients with suspected or confirmed septic shock, immediate empiric antimicrobial therapy within 1 h of shock recognition is recommended. The association between time to antibiotics and mortality in patients with suspected or confirmed sepsis without shock has not been established. In patients in whom a diagnosis of sepsis is less certain and shock is absent, further time-limited clinical evaluation is recommended prior to empiric antibiotic administration. If an alternative diagnosis is not identified within 3 h of clinical presentation, empiric antibiotic administration is recommended.

Selection of initial empiric antibiotics should consider site of infection and potential etiologic organisms; community versus health care exposure; known prior infections, antibiotic usage, and local antimicrobial resistance profiles; and the patient's immune status, comorbidities, and illness severity. Recommendations for initial empiric antibiotic use based on site of infection and other pertinent factors are summarized in **Table 315-1**. In patients with undifferentiated sepsis, in whom the primary site of infection is unclear, use of broad-spectrum antibiotics with a high likelihood of in vitro susceptibility to all organisms likely causing infection is recommended. In patients in whom *Pseudomonas* is not considered a likely pathogen, then a third-generation cephalosporin such as ceftriaxone or cefotaxime is recommended for gram-negative bacteria coverage. If *Pseudomonas* is likely, then cefepime, piperacillin-tazobactam, or a carbapenem such as imipenem or meropenem is recommended for gram-negative bacteria coverage. In patients at risk for highly resistant gram-negative infections (e.g., patients with prior known highly resistant infections or colonization), use of two empiric gram-negative antibiotics is recommended. Finally, in patients with undifferentiated sepsis with risk factors for MRSA (e.g., frequent health care exposure or hospital-onset sepsis), vancomycin or linezolid administration is recommended.

Optimization of antibiotic delivery, such as administering β-lactam antibiotics prior to vancomycin, prolonged infusion of β-lactam antibiotics after the initial infusion, and optimization of pharmacokinetics/pharmacodynamics, should be considered in consultation with trained pharmacy and infectious diseases experts. Routine empiric antifungal therapy use in patients with undifferentiated sepsis is not recommended. However, in patients at increased risk of fungal infection (e.g., recent abdominal surgery, parenteral nutrition, liver failure, diabetes, colonization of multiple anatomic sites with *Candida* spp.), empiric echinocandin administration is recommended. While the Surviving Sepsis Campaign does not provide recommendations on antiviral use, remdesivir or neuraminidase inhibitor (e.g., oseltamivir) administration in septic patients with severe acute respiratory syndrome coronavirus-2 (SARS-CoV-2) and influenza infection, respectively, should be considered.

In addition to providing appropriate and timely antibiotics, identifying sources of infection amendable to control is imperative. Examples include, but are not limited to, intraabdominal abscess, bowel perforation, pyelonephritis, cholangitis, and necrotizing skin and soft tissue infections that are amendable to surgical or procedural source control. Source control should occur as rapidly as possible. Furthermore, removal of indwelling catheters should occur if the catheter appears infected (e.g., erythema or purulence at the catheter insertion site), if endovascular infection (e.g., endocarditis) is suspected or documented, and in critically ill patients without a clear alternate source of infection. Finally, antibiotic stewardship to limit development of resistant organisms and other antibiotic-associated complications (e.g., *Clostridioides difficile* infection, hypersensitivity/allergic reaction, and acute kidney

TABLE 315-1 Site-Specific Empiric Antibiotic Recommendations

SITE OF INFECTION	INITIAL EMPIRIC THERAPY	OTHER CONSIDERATIONS
Pulmonary		
CAP	Multidrug therapy with a β-lactam (ampicillin + sulbactam, ceftriaxone, or ceftaroline) and a macrolide (azithromycin or clarithromycin) Monotherapy with a respiratory fluoroquinolone (levofloxacin or moxifloxacin)	Risk factors for MRSA and/or *Pseudomonas aeruginosa*: add vancomycin or linezolid for MRSA coverage; replace standard CAP therapy with antipseudomonal coverage such as piperacillin-tazobactam, cefepime, meropenem, or imipenem. Recommendation based on "local validation" of risk factors for community-onset MRSA or *P. aeruginosa* or prior isolation of these organisms in the previous year, particularly from respiratory specimens.
HAP/VAP	Multidrug therapy with vancomycin or linezolid and piperacillin-tazobactam, cefepime, ceftazidime, imipenem, meropenem, or aztreonam	Two antipseudomonal antibiotics from different classes (addition of fluoroquinolones, aminoglycosides, or polymyxins) if prior intravenous antibiotic use within 90 days for HAP/VAP and septic shock at time of VAP, ARDS preceding VAP, 5 or more days of hospitalization prior to VAP, or acute renal replacement therapy prior to VAP. If prior colonization with carbapenem-resistant Enterobacterales or KPC-producing organism, ceftazidime-avibactam and meropenem-vaborbactam should be considered, but further efficacy data are needed. Empiric regimens should be informed by local distribution of pathogens and their antimicrobial susceptibilities.
Central Nervous System		
Health care–associated ventriculitis and meningitis	Vancomycin and cefepime, ceftazidime, or meropenem	β-Lactam choice based on local in vitro susceptibility patterns. If carbapenem-resistant *Acinetobacter* is suspected, addition of meropenem and colistin or polymyxin B.
Meningitis	Vancomycin and ceftriaxone	Age >50, alcohol abuse, or immunocompromised: add ampicillin. Penetrating head trauma, CSF shunt, or postneurosurgery, vancomycin and cefepime, ceftazidime, or meropenem. Clinical presentation suggestive of *Rickettsia* or *Ehrlichia*, add doxycycline.
Skin and Soft Tissue		
Necrotizing fasciitis including Fournier gangrene	Multidrug therapy with vancomycin or linezolid and piperacillin-tazobactam, a carbapenem, or ceftriaxone and metronidazole	Prompt surgical consultation is recommended for patients with aggressive infections associated with signs of systemic toxicity or suspicion of necrotizing fasciitis or gas gangrene.
Nonpurulent cellulitis/ erysipelas (severe)	Vancomycin and piperacillin-tazobactam	Emergent surgical inspection to rule out necrotizing process.
Purulent furuncle/ carbuncle/abscess (severe)	Vancomycin, daptomycin, linezolid, telavancin, or ceftaroline	Incision and drainage as indicated.
Intraabdominal		
Community-onset extrabiliary (mild)	Cefoxitin, ertapenem, moxifloxacin, or tigecycline	Health care setting with high prevalence of ESBL-producing Enterobacterales or >20% of *Pseudomonas* resistant to ceftazidime, consider carbapenem or piperacillin-tazobactam.
Community-onset extrabiliary (severe)	Imipenem-cilastatin, meropenem, doripenem, or piperacillin-tazobactam	Health care associated: imipenem-cilastatin, meropenem, or piperacillin-tazobactam, levofloxacin, or cefepime each along with metronidazole, vancomycin added to each regimen.
Community-onset biliary (mild to moderate)	Cefazolin, cefuroxime, or ceftriaxone	Empiric therapy should be driven by local microbiologic data and source control performed as indicated.
Community-onset biliary (severe) or cholangitis	Imipenem-cilastatin, meropenem, or piperacillin-tazobactam, levofloxacin, or cefepime each in combination with metronidazole	
Genitourinary		
Acute pyelonephritis (IDSA archived)	Ceftriaxone, trimethoprim-sulfamethoxazole, or ciprofloxacin	Requiring hospitalization: intravenous fluoroquinolone, aminoglycoside, extended-spectrum cephalosporin, extended-spectrum penicillin, or carbapenem with choice of agents based on local resistance data. Do not use fluoroquinolone if >10% resistance prevalence or trimethoprim-sulfamethoxazole in areas of high resistance.

Abbreviations: ARDS acute respiratory distress syndrome; CAP, community acquired pneumonia; CSF, cerebrospinal fluid; HAP, hospital-acquired pneumonia; IDSA, Infectious Diseases Society of America; KPC, *Klebsiella pneumoniae* carbapenemase; ESBL, extended spectrum beta-lactamase; MRSA, methicillin-resistant *Staphylococcus aureus*; VAP, ventilator-associated pneumonia.

injury) requires regular reassessment of the patient's clinical course and available microbiologic data to guide narrowing the spectrum of antibiotic therapy and using shorter rather than longer courses of therapy.

NEUROLOGIC COMPLICATIONS

Encephalopathy, manifested by altered consciousness, cognition, or attention, ranging from mild delirium to coma, that is not attributable to an alternative etiology, is the most common neurologic complication of sepsis (Table 315–2). Sepsis-associated encephalopathy occurs in over half of septic patients and is associated with increased mortality and long-term functional and neuropsychiatric sequalae among survivors. Septic patients may also experience hyper- or hypoactive delirium, seizure, stoke, central nervous system infection, or coma, not directly attributable to sepsis. Thus, further diagnostic evaluation is needed in septic patients with clinical findings indicative of seizures (i.e., electroencephalogram), stroke (i.e., brain imaging), or meningoencephalitis (i.e., lumbar puncture) to identify and treat these complications. Underlying factors contributing to sepsis-associated encephalopathy include decreased cerebral perfusion, microcirculatory and blood-brain barrier disruption, and exposure of brain parenchyma to circulating inflammatory mediators and oxidative stress resulting in neuronal injury, dysfunction, and death. Neuronal and glial apoptosis

TABLE 315-2 Organ-Specific Clinical Findings and Management

	EPIDEMIOLOGY	CLINICAL FINDINGS	DIAGNOSIS	TREATMENT
Neurologic	54% of septic patients develop encephalopathy	Altered consciousness, cognition, or attention; seizure, stroke, or meningism	EEG, brain imaging (MRI or CT), lumbar puncture	Early recognition and supportive care; treat underlying cause; correct metabolic and electrolyte abnormalities; limit neurotoxic agents and sedatives
Cardiovascular	25% of septic patients develop shock and half have myocardial dysfunction	Tachycardia, hypotension, skin mottling, prolonged capillary refill, oliguria, altered mental status	Invasive blood pressure monitoring, dynamic assessment of volume status, echocardiogram	Intravenous fluid resuscitation with balanced crystalloid, ~30 mL/kg; vasopressors for persistent hypotension, norepinephrine (first agent), vasopressin (second agent), epinephrine (third agent); hydrocortisone 200 mg/d if ongoing vasopressor requirement; consider adding dobutamine to norepinephrine or switch to epinephrine in septic shock patients with decreased LVEF; consider use of pulmonary artery catheter in patients with mixed septic and cardiogenic shock
Respiratory	7% of septic patients develop ARDS	Tachypnea, hypoxia, increased work of breathing	Chest x-ray or ultrasound with noncardiogenic bilateral infiltrates and Pa_{O_2}/Fi_{O_2} <300 mmHg or Sp_{O_2}/Fi_{O_2} ≤315	Maintain Sp_{O_2} 90–96%; use high-flow nasal canula in patients with adequate neurologic status; target plateau pressure <30 cmH_2O and tidal volume of 6–8 mL; consider rescue therapies in patients with refractory hypoxia
Genitourinary	67% of septic patients have acute kidney injury	Oligura or anuria	Elevated serum creatinine and blood urea nitrogen, acidemia, hyperkalemia	Treat underlying infection, maintain renal perfusion, avoid nephrotoxic agents; start renal replacement therapy for progressive acidemia, hyperkalemia, uremia, or volume overload; infuse sodium bicarbonate if renal failure and pH ≤7.2.
Gastrointestinal	50% of septic shock patients have liver dysfunction	Right upper quadrant pain, asterixis, jaundice	Elevated bilirubin, alkaline phosphatase, and transaminases; right upper quadrant ultrasound	Treat underlying infection, avoid hypotension and hepatoxic agents; stress ulcer prophylaxis for high-risk patients; enteral feeding if shock controlled within 48 h; parenteral feeding if nutrition goal not met within 7 days; insulin if blood glucose ≥180 g/dL
Hematologic	35% of septic shock patients have disseminated intravascular coagulation	Clinical or subclinical thrombosis or hemorrhage	Thrombocytopenia, increased fibrin split products, decreased fibrinogen, prolonged PT/INR	Administer cryoprecipitate for fibrinogen <150 mg/dL, platelets for platelet count ≤50×10^9/L and evidence of bleeding, and fresh frozen plasma for prolongated PT/INR and evidence of bleeding; transfuse packed red blood cells for hemoglobin <7.0 g/dL

Abbreviations: ARDS, acute respiratory distress syndrome; CT, computed tomography; EEG, electroencephalogram; INR, international normalized ratio; LVEF, left ventricular ejection fraction; MRI, magnetic resonance imaging; PT, prothrombin time.

are correlative findings at autopsy among fatal cases. The management of sepsis-associated encephalopathy includes early recognition, treatment of the underlying infection, supportive care including correction of metabolic or electrolyte abnormalities, and limiting exposure to pharmacologic agents that are neurotoxic (e.g., cefepime) or have central nervous system effects including opiates and benzodiazepines that might exacerbate the condition or predispose to long-term disability or neurocognitive dysfunction.

CARDIOVASCULAR DYSFUNCTION

Cardiovascular comprise, manifested as hypotension or shock, occurs in ~25% of septic patients. Hypotension results from peripheral arteriolar vasodilation and decreased cardiac venous return due to venous vasodilation and intra- to extravascular fluid shifts. Clinical signs of diminished tissue perfusion include skin mottling, prolonged capillary refill time, oliguria, and altered mental status. During early compensated shock, heart rate and cardiac output increase. As septic shock progresses, loss of vascular smooth muscle contractility, despite endogenous neurohormonal stimuli and exogenous catecholamine administration, results in progressive or refractory shock.

Up to half of patients with septic shock also have myocardial dysfunction, which is associated with increased mortality. Sepsis-induced cardiomyopathy manifests as decreased left ventricular ejection fraction, increased end-diastolic volume index, and right ventricular impairment. Contributing factors to sepsis-induced cardiomyopathy include global ischemia, hypoxia, and impaired myocardial metabolism; endothelial damage, increased adhesion molecule expression, and coronary microcirculatory dysfunction; and direct cardiomyocyte suppression, mitochondrial dysfunction, or cardiomyocyte death from exposure to inflammatory mediators including cytokines (e.g., IL-1β, IL-6, and TNF-α), complement proteins, and NO.

Intravenously administered fluids and vasopressors are used to restore and maintain blood pressure and tissue perfusion in the setting of septic shock. An indwelling arterial catheter should be placed as soon as feasible to continuously monitor blood pressure invasively. Crystalloid solutions are preferred over colloid for initial volume resuscitation, although colloid such as albumin can be considered in patients who have already received large volumes of crystalloid. Balanced crystalloid solution, such as lactated Ringer's, may be preferrable to 0.9% normal saline, which is more likely to induce hyperchloremic metabolic acidosis associated with renal vasoconstriction and kidney injury. Use of pentastarch or hydroxyethyl starch is associated with severe kidney injury and death and should be avoided.

Crystalloid should be administered as a bolus over 5–10 min at a volume of ~30 mL/kg. However, 30 mL/kg may not be appropriate for all patients, including those with end-stage renal disease and systolic heart failure. Protocolized fluid resuscitation targets including central venous pressure (CVP) of 8–12 mmHg, central venous oxygen saturation >70%, and urine output ≥0.5 mL/kg per h have not been associated with improved mortality and so are not recommended to guide total fluid replacement. Instead, dynamic assessment of volume responsiveness using capillary refill time, passive leg raise maneuver, and point-of-care ultrasound, with iterative reevaluation, should guide total volume administered and is preferred over static measurements of volume responsiveness such as CVP. Notably, a 2017–2018 clinical trial conducted in 28 ICUs in five countries found that resuscitation guided by capillary refill time compared with lactate level–targeted resuscitation did not reduce all-cause 28-day mortality, an important finding for sepsis care in resource-limited settings.

In patients who have received adequate fluid resuscitation yet remain hypotensive, a continuous norepinephrine drip should be initiated as

the first-line vasopressor to maintain a mean arterial pressure target of ≥65 mmHg. Vasopressin should be added as a second agent, and epinephrine as a third agent, if needed, to achieve the blood pressure target. A contemporary randomized clinical trial tested the efficacy of early restrictive (i.e., prioritizing vasopressors and lower intravenous fluid volumes) versus liberal (i.e., prioritizing higher volumes of intravenous fluids before vasopressors use) fluid management in the first 4 h of presentation in patients with septic shock. Across 1563 patients at 60 U.S. medical centers, there was no difference in 90-day mortality. In septic patients with evidence of myocardial dysfunction (e.g., low cardiac output, elevated filling pressures) and evidence of persistent hypoperfusion following adequate volume resuscitation, consideration should be given to adding dobutamine to norepinephrine or using epinephrine alone to increase inotropy. While routine use of pulmonary artery catheters to guide fluid management in sepsis has not been associated with improved outcomes, use may be considered in patients with mixed septic and cardiogenic shock.

In patients with septic shock and ongoing requirement for vasopressor therapy, it is recommended to start intravenous corticosteroids with hydrocortisone at a dose of 200 mg/d often provided as 50 mg every 6 h.

ACUTE LUNG INJURY

Lung injury, manifesting as acute hypoxic respiratory insufficiency or failure, and termed acute respiratory distress syndrome (ARDS), complicates ~7% of sepsis cases. ARDS is a syndrome that is classified by noncardiogenic diffuse pulmonary infiltrates and hypoxemia, in which infiltrates can be determined by x-ray or ultrasound imaging, and hypoxemia can be determined by Pa_{O_2}/F_{IO_2} <300 mmHg or percent oxygen saturation (Sp_{O_2})/F_{IO_2} ≤315. The pathogenesis of ARDS overlaps with that of sepsis, in which activated myeloid cells and soluble inflammatory mediators disrupt pneumocytes' and endothelial cells' structure and function, resulting in leakage of plasma components and further recruitment of immune cells into lung alveolar and interstitial spaces. The histopathologic correlate of ARDS is diffuse alveolar damage, which progresses through early exudative, then proliferative, and finally fibrotic stages.

Initial management of acute lung injury in sepsis requires administration of oxygen to meet cellular metabolic demand while limiting cellular injury from oxidative stress, with a reasonable Sp_{O_2} target being 90–96%. In patients with adequate neurologic status, and absent other specific contraindications, high-flow nasal canula is preferred over noninvasive ventilation to improve hypoxia in patients without hypercapnia. For patients with ARDS who require invasive mechanical ventilation, targeting a plateau pressure of <30 cmH_2O and a tidal volume of 6–8 mL/kg, based on ideal body weight, has been associated with reduced mortality and is recommended. In patients with refractory hypoxia despite low tidal volume ventilation, rescue measures including prone positioning, neuromuscular blockade, and venovenous extracorporeal membrane oxygenation should be considered.

ACUTE KIDNEY INJURY

Acute kidney injury (AKI) occurs in up to two-thirds of patients with sepsis or septic shock and is associated with increased mortality and risk of chronic kidney disease and disability among survivors. Sepsis-associated AKI presents with oliguria or anuria and elevated serum creatinine and blood urea nitrogen and accounts for all AKI cases occurring within 7 days of sepsis onset. Sepsis-associated AKI may be attributed to sepsis, associated complications (e.g., abdominal compartment syndrome), and clinical management including administration of nephrotoxic substances such as antibiotics and intravenous contrast dye. Factors that contribute to pathogenesis of AKI due to sepsis alone, termed sepsis-induced AKI, include renal hypoperfusion, microvascular injury and dysfunction, inflammatory cellular and soluble mediators, and altered renal tubular metabolic and mitochondrial function. Many serum and urine biomarkers have been evaluated to improve early recognition and treatment of sepsis-associated AKI. However, to date, none have been shown to improve outcomes.

Management of sepsis-associated AKI includes treating underlying infection, maintaining renal perfusion with fluid resuscitation and vasopressors to achieve blood pressure goals, avoiding nephrotoxic agents when possible, and identifying and treating reversible causes. Initiating early renal replacement therapy (RRT), including intermittent hemodialysis or continuous renal replacement therapy (CRRT), during sepsis-associated AKI is not associated with improved outcomes, and so RRT should be started for standard definitive indications including progressive acidemia, hyperkalemia, uremia, or volume overload. CRRT is better tolerated than intermittent hemodialysis in patients with septic shock, allowing for less dynamic intravascular volume shifts and so is preferred in these patients when available. Sodium bicarbonate infusion should be considered only in patients with AKI and severe metabolic acidemia (pH ≤7.2) to maintain vasopressor effectiveness and mitigate fatal ventricular arrythmia risk.

GASTROINTESTINAL COMPLICATIONS

Approximately 50% of patients with septic shock develop liver dysfunction, which is associated with increased mortality. Clinical findings may include right upper quadrant pain, jaundice, and asterixis depending on severity. Common laboratory findings include elevated serum bilirubin and alkaline phosphatase levels and elevated transaminases if marked hypotension occurs. Sepsis-induced cholestasis is attributed to impaired bile formation and decreased flow in a nonobstructive pattern. The liver plays an essential role in microbial clearance, in which the Kupffer cells phagocytose bacteria, release proinflammatory cytokines and chemokines, and bind platelets. Neutrophils are recruited to liver sinusoids, release NETs to trap pathogens, and contribute to a proinflammatory and prothrombotic environment. In autopsy studies of fatal sepsis, hepatitis and steatosis are detected in most patients, while portal inflammation, centrilobular necrosis, hepatocellular apoptosis, and cholangitis may also be detected. Management of liver injury and dysfunction during sepsis includes treating the underlying infection and avoiding hypotension and hepatoxic medications. In septic patients with elevated serum bilirubin, abdominal ultrasound should be performed to evaluate for biliary obstruction, cholecystitis, and cholangitis.

Sepsis has also been associated with alteration of the intestinal mucosa including increased epithelial permeability, changes in gut microbiota, and translocation of enteric microbes into circulation. In patients with sepsis and septic shock and risk factors for gastrointestinal bleeding (e.g., mechanical ventilation, coagulopathy, preexisting liver disease, high organ failure score, and need for RRT), stress ulcer prophylaxis is recommended. Decreased splanchnic perfusion and increased interstitial edema contribute to impaired intestinal absorption, which in combination with the catabolism of sepsis contributes to nutritional deficiencies. The route and timing of supplemental nutrition administration in patients with sepsis and septic shock are debated. Current recommendations are to initiate enteral nutrition early, within 48 h of diagnosis, in patients in whom shock is controlled with fluids and vasopressors and absent contraindications like bowel ischemia. The rationale for early enteral feeding is to maintain enteric epithelial integrity while limiting negative nitrogen and caloric balance. Supplemental use of parenteral nutrition is suggested if nutritional goals are unmet within 7 days by enteric feeding and supplemental glucose infusion. Insulin therapy is recommended for patients with blood glucose levels >180 mg/dL.

HEMATOLOGIC COMPLICATIONS

Coagulation abnormalities in sepsis are common, ranging from isolated thrombocytopenia to disseminated intravascular coagulation (DIC), and can manifest with clinical or subclinical thrombosis and hemorrhage. Up to 35% of septic shock patients meet DIC criteria, including thrombocytopenia, increased fibrin split products, decreased fibrinogen, and prolonged prothrombin time (PT)/international normalized ratio (INR), which is associated with increased mortality. Given that DIC is a late manifestation of coagulation abnormalities in sepsis, a scoring system for earlier detection of sepsis-induced coagulopathy (SIC) has been proposed. SIC scoring, which considers platelet counts, PT, and SOFA score, is more sensitive than DIC criteria for recognizing coagulation abnormalities in sepsis. However, the clinical utility of SIC